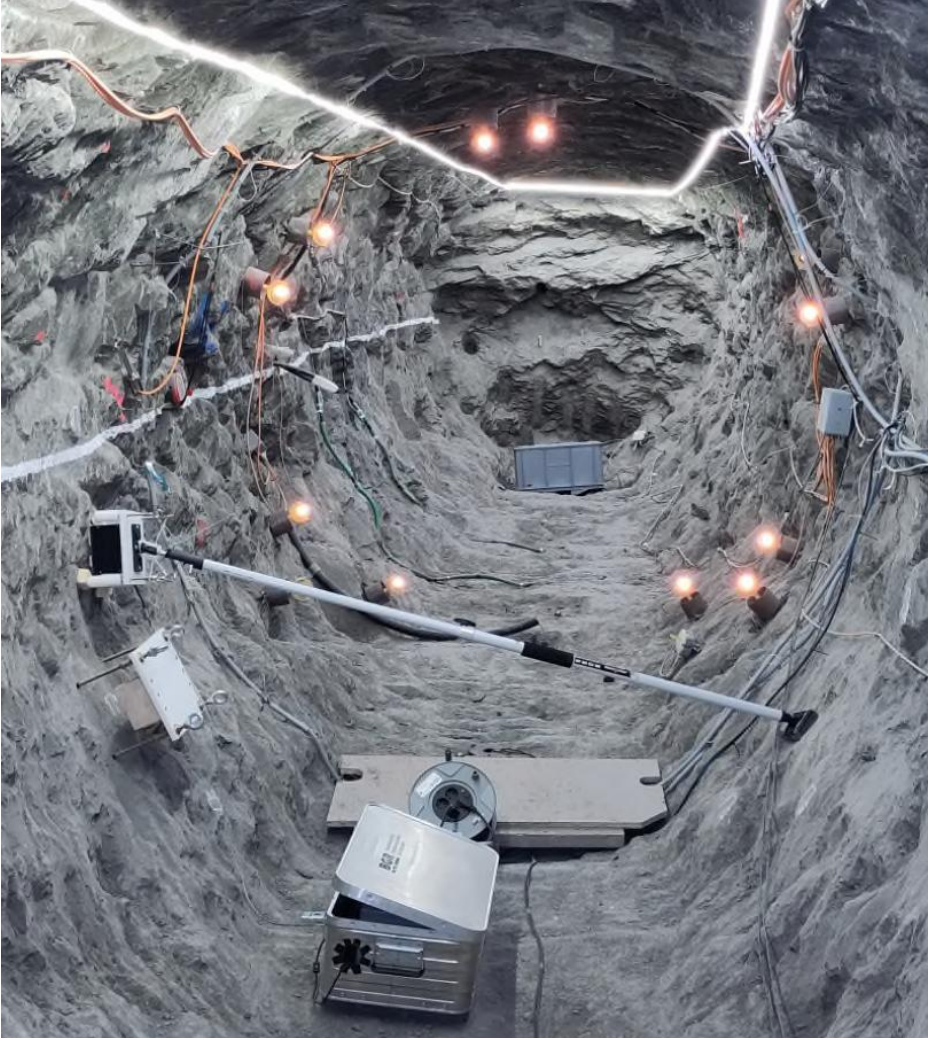
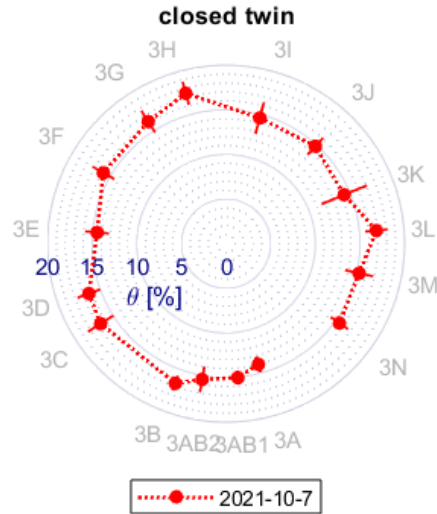
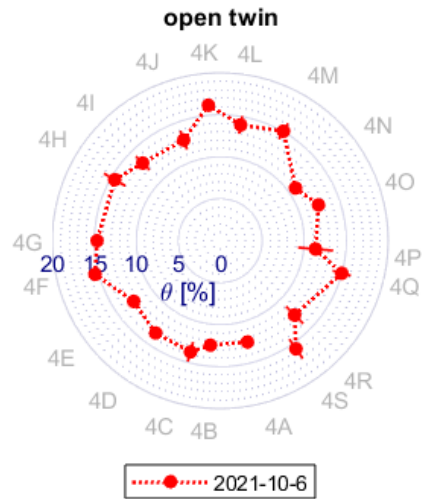


Single-Sided Nuclear Magnetic Resonance on the walls in the CD-A Niches

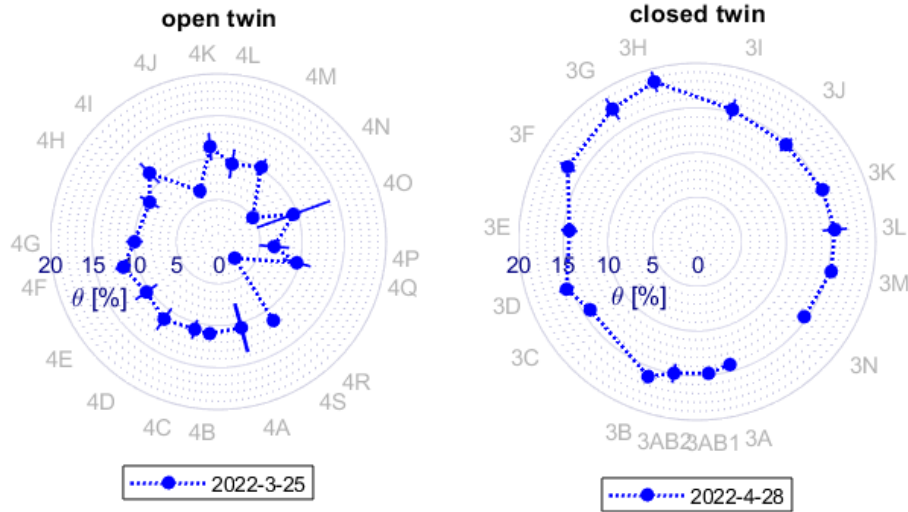
Stephan Costabel, BGR, working group B2.1



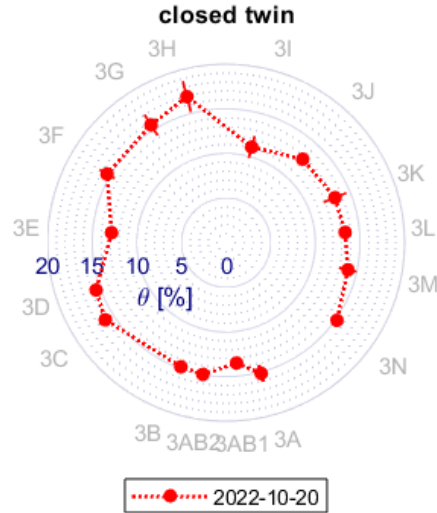
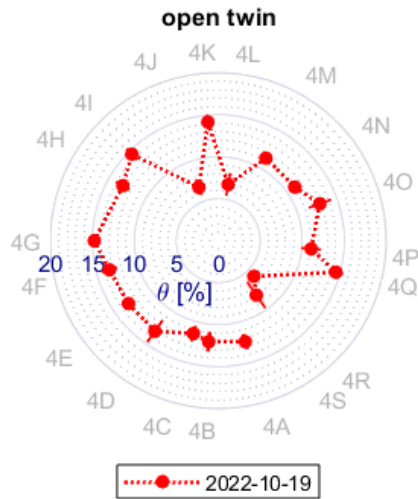
Seasonal measurements in both niches - data until Feb 2025



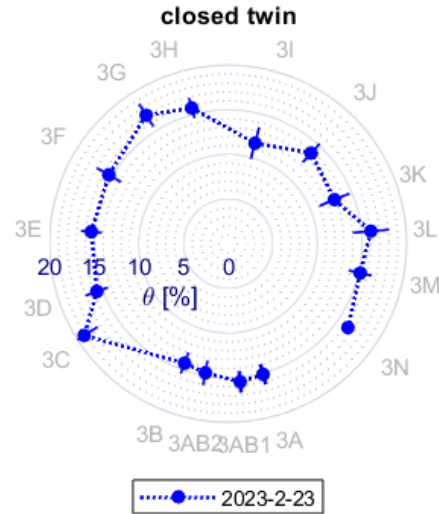
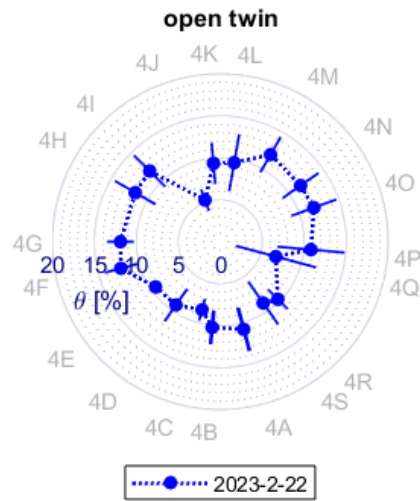
Seasonal measurements in both niches - data until Feb 2025



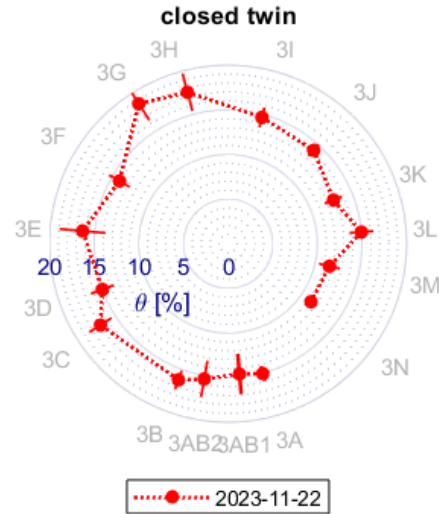
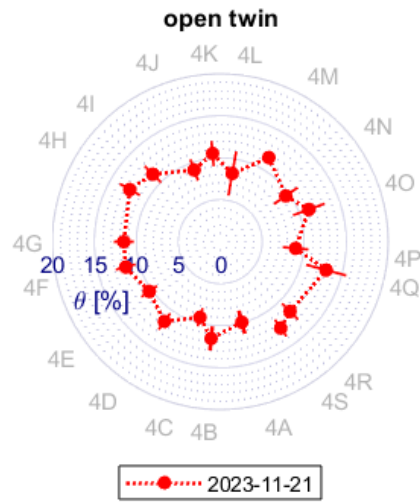
Seasonal measurements in both niches - data until Feb 2025



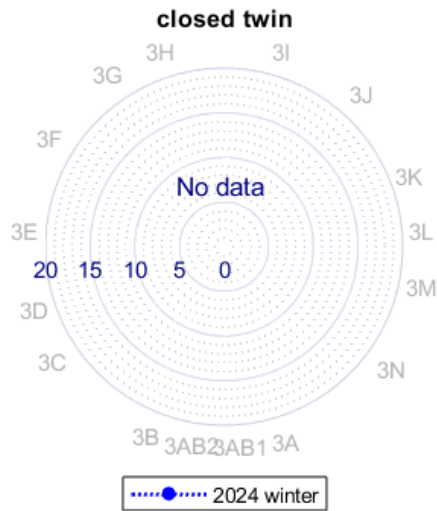
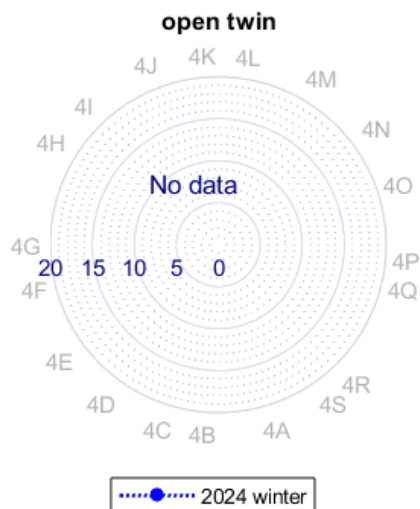
Seasonal measurements in both niches - data until Feb 2025



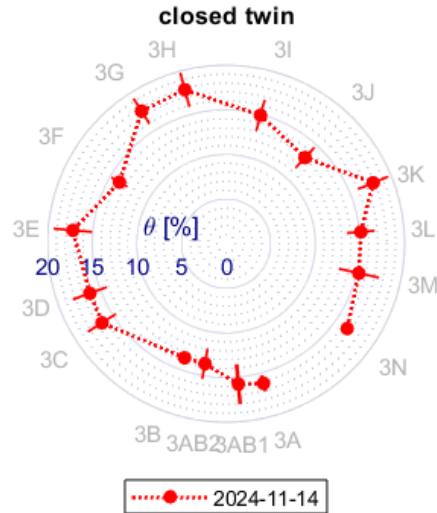
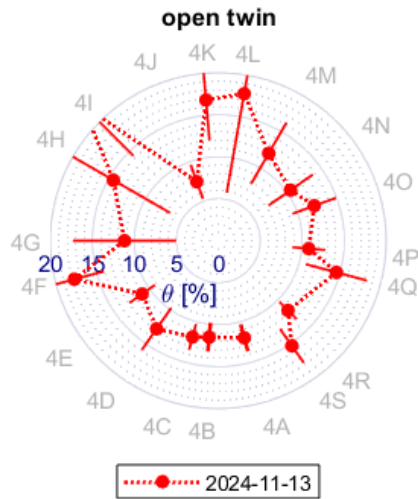
Seasonal measurements in both niches - data until Feb 2025



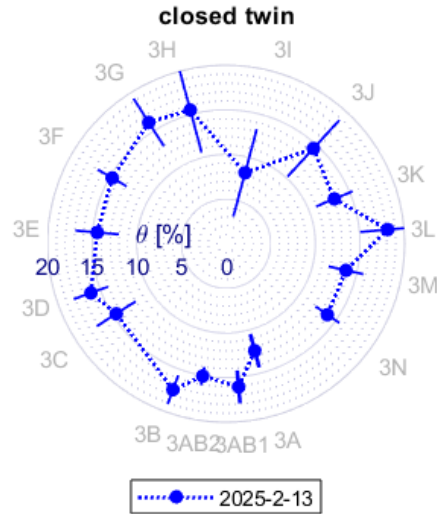
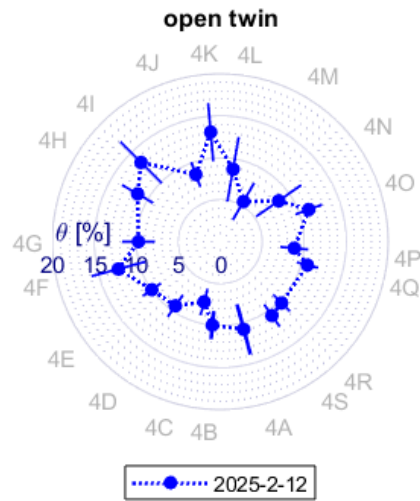
Seasonal measurements in both niches - data until Feb 2025



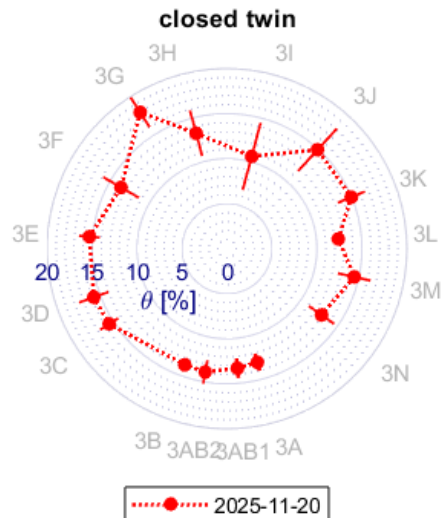
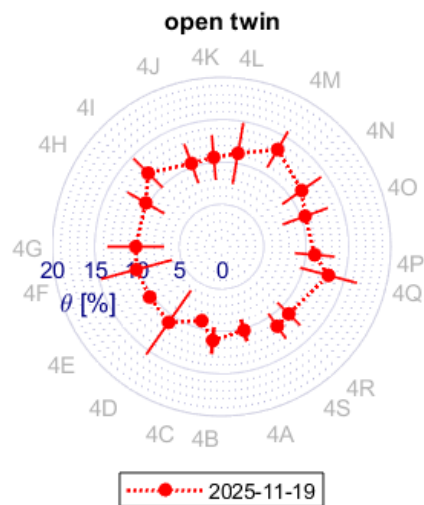
Seasonal measurements in both niches - data until Feb 2025



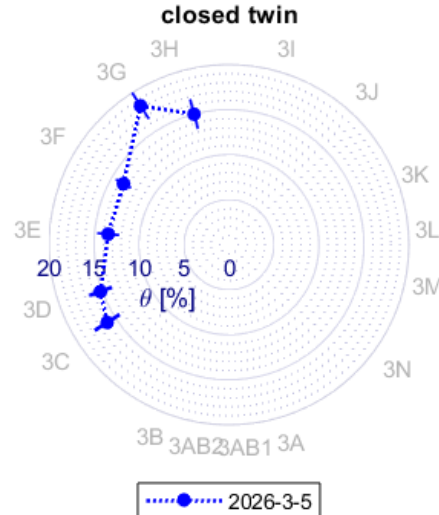
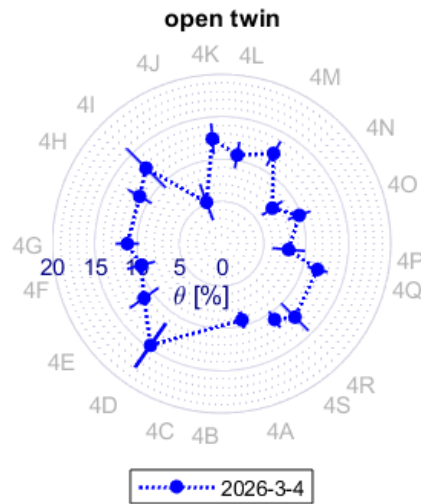
Seasonal measurements in both niches - data until Feb 2025



Seasonal measurements in both niches - data until Feb 2025

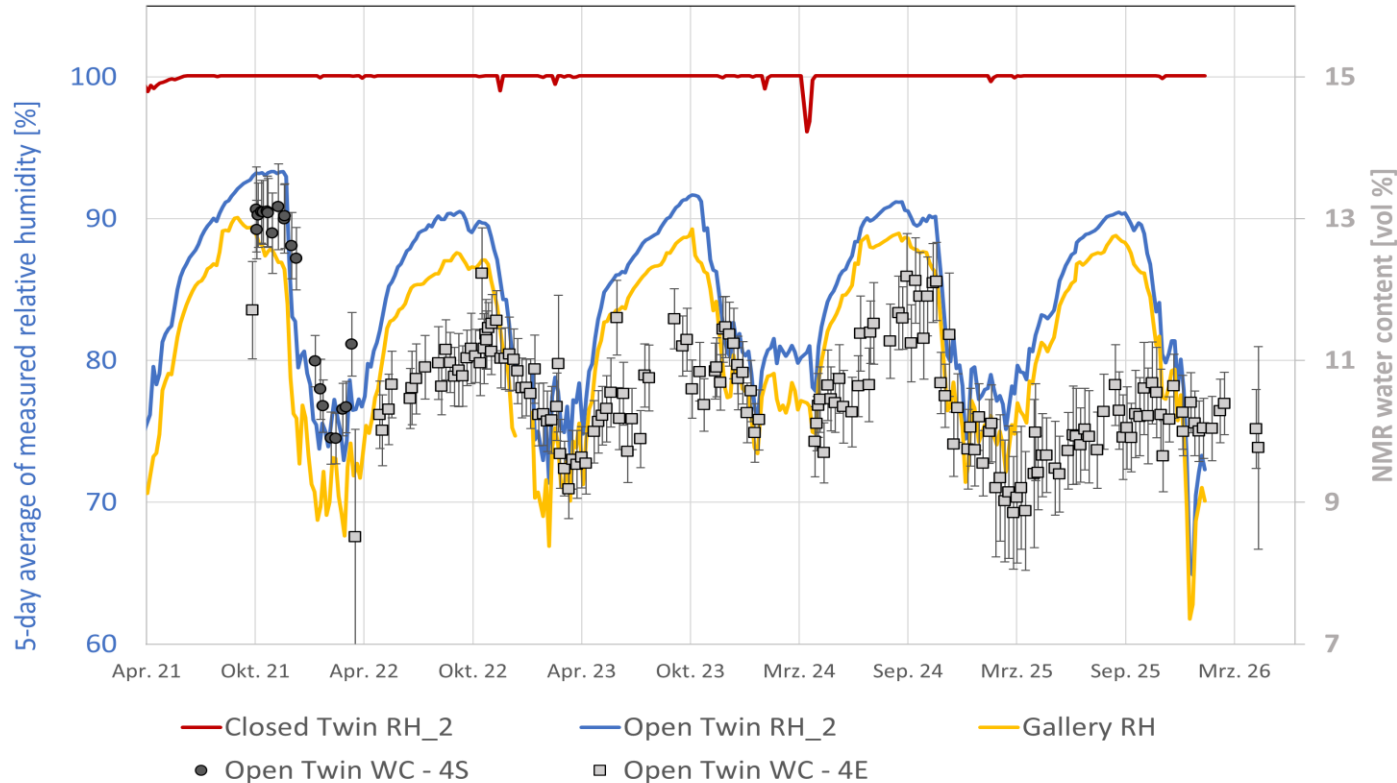


Seasonal measurements in both niches - data until Feb 2025



- **Closed niche:** behaves as expected - no significant difference
- **Open niche:** seasonal changes seem to be decreasing !?

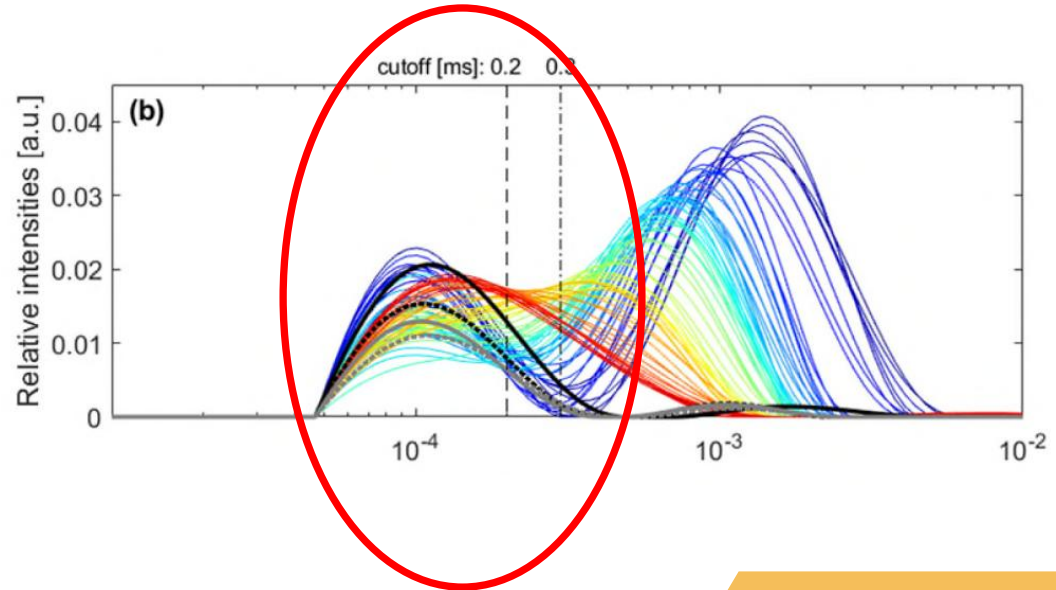
Weekly measurements in open niche - relationship with rel. humidity (apparent phase shift!?)

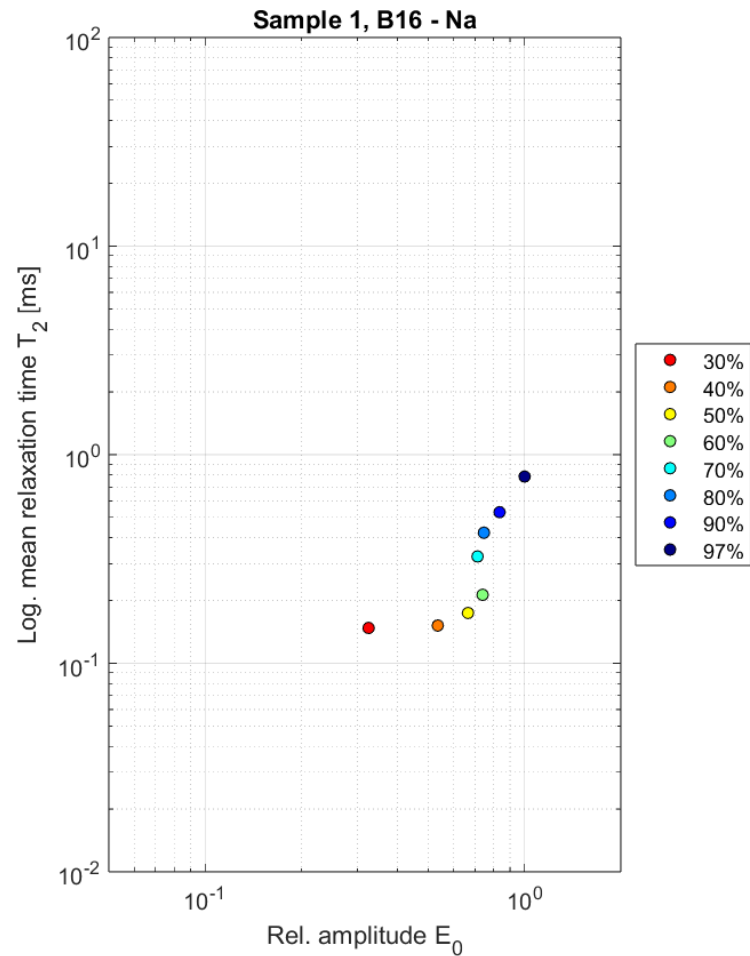
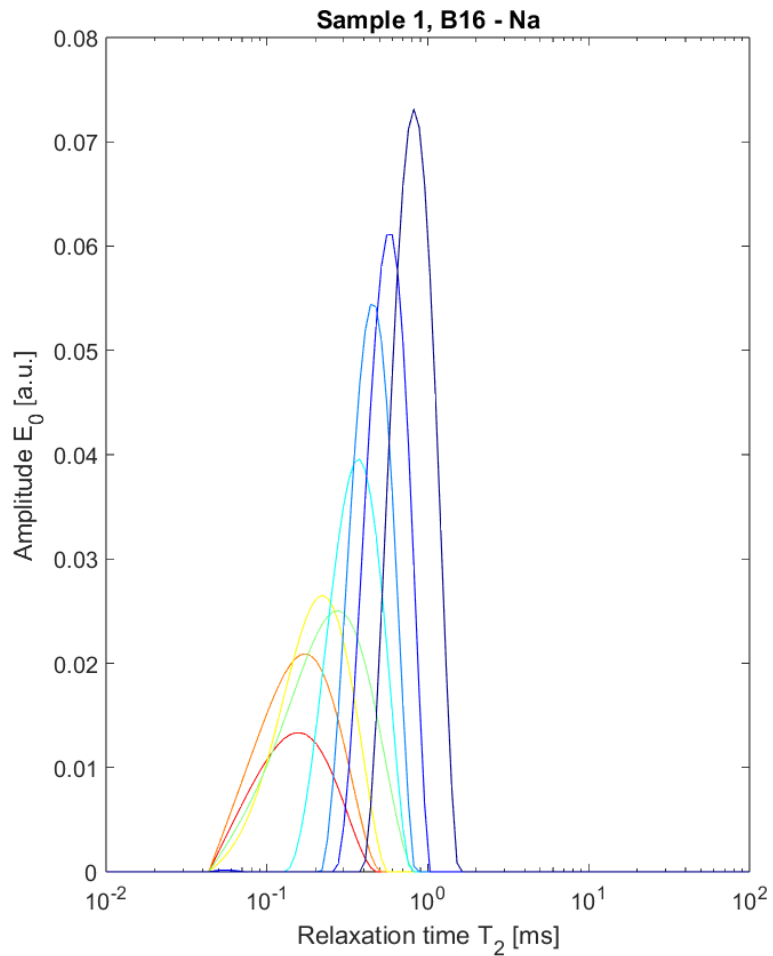


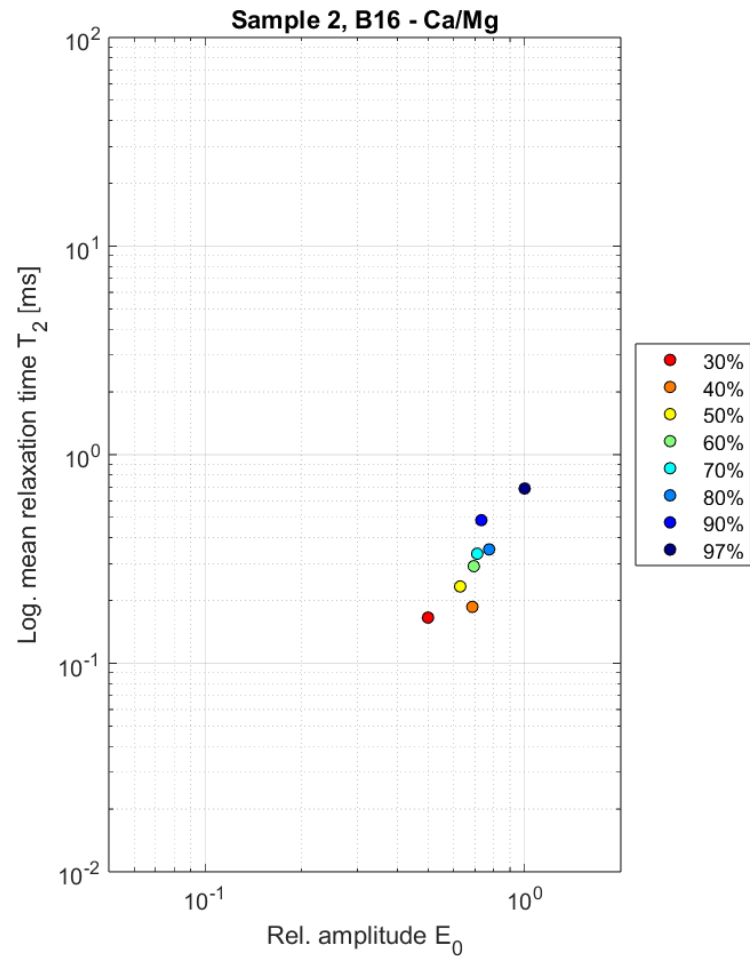
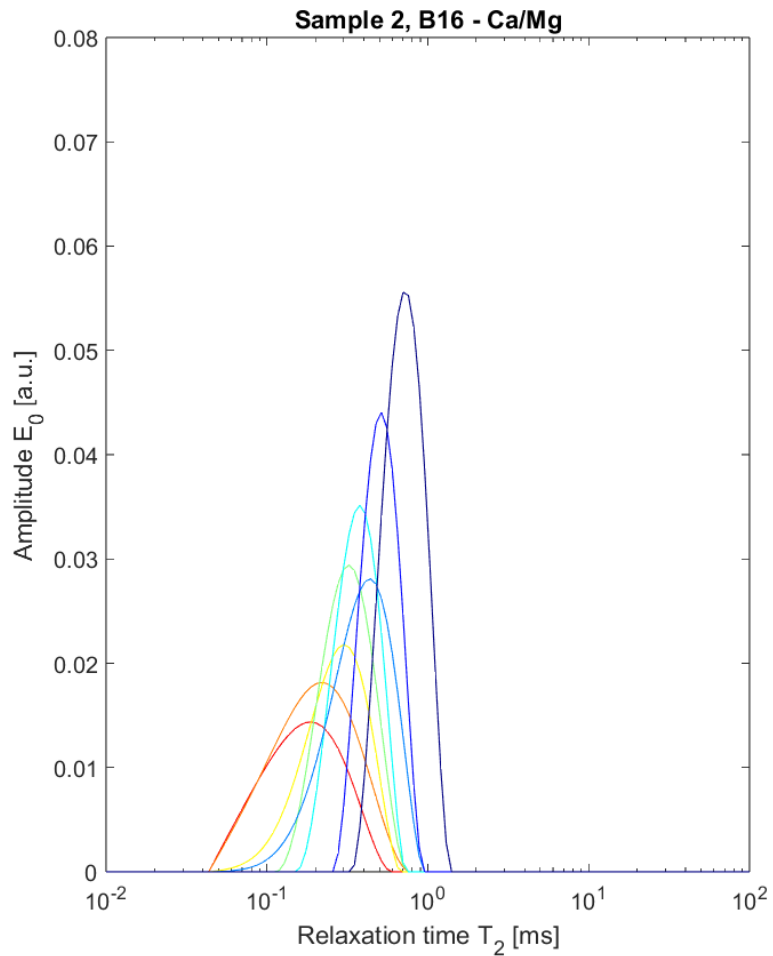
- Closed niche: behaves as expected - no significant difference
- Open niche: seasonal changes seem to be decreasing !?

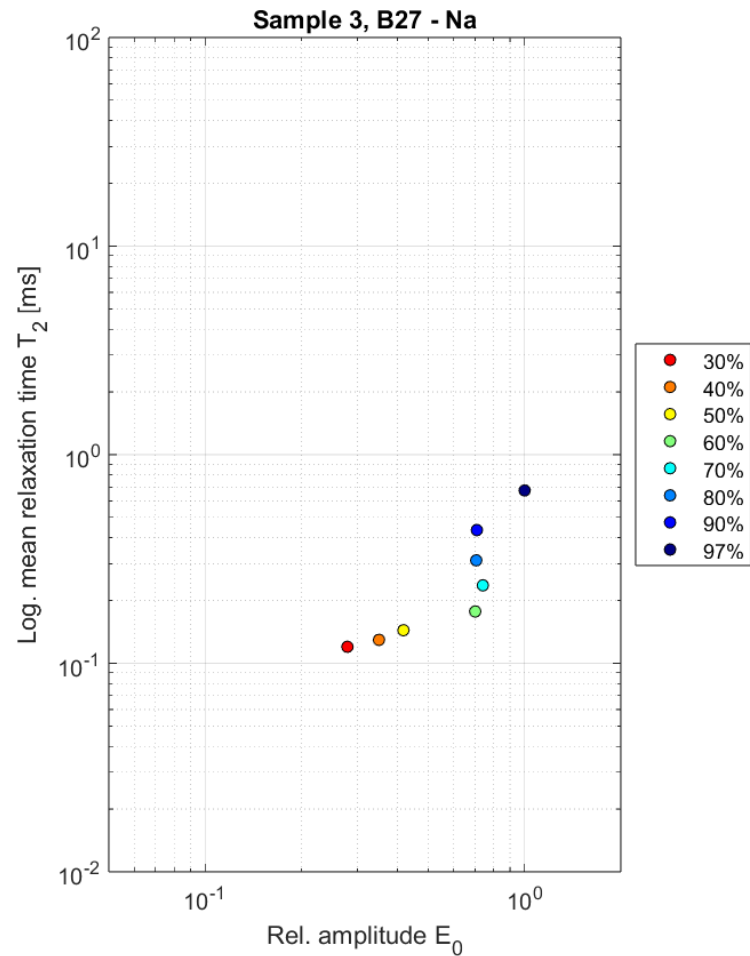
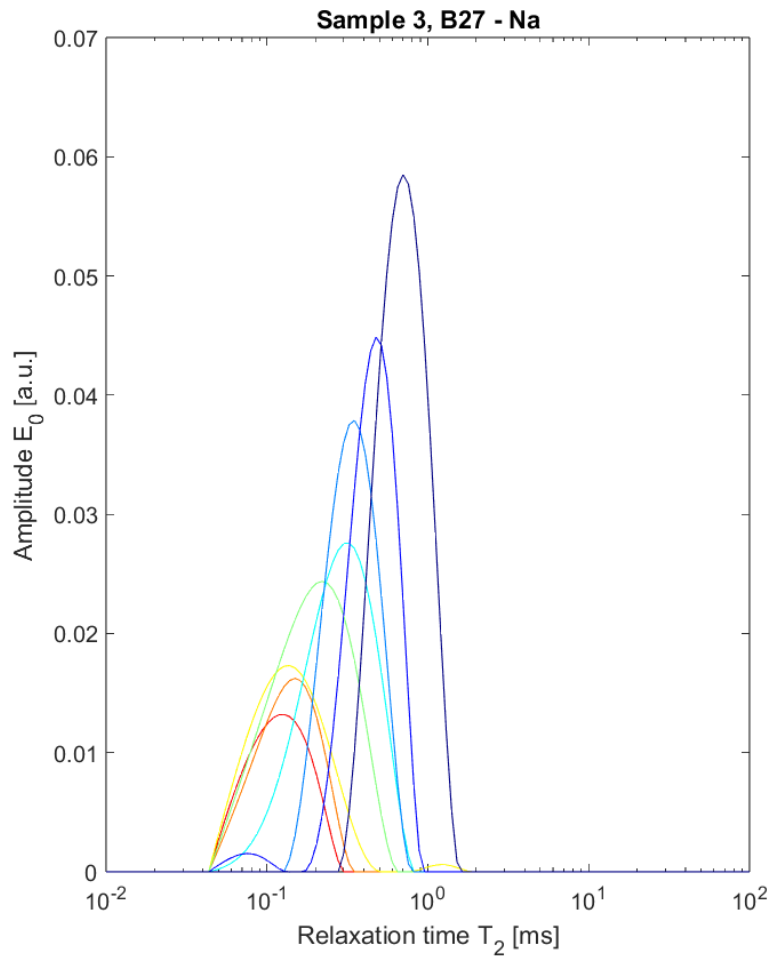
Laboratory analysis on clay mineral powders

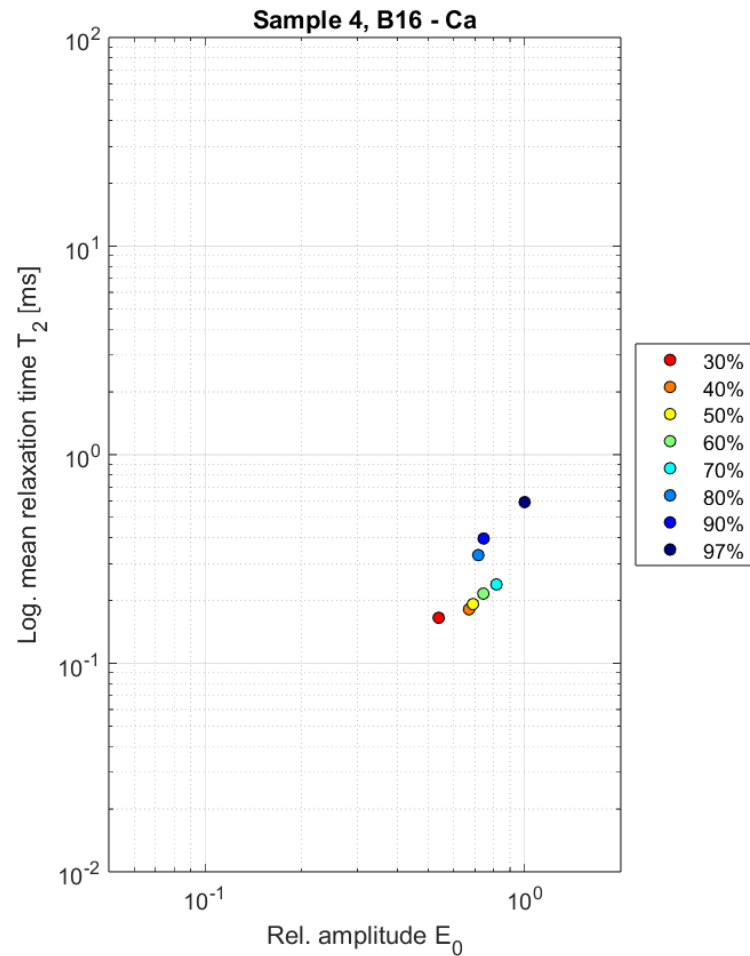
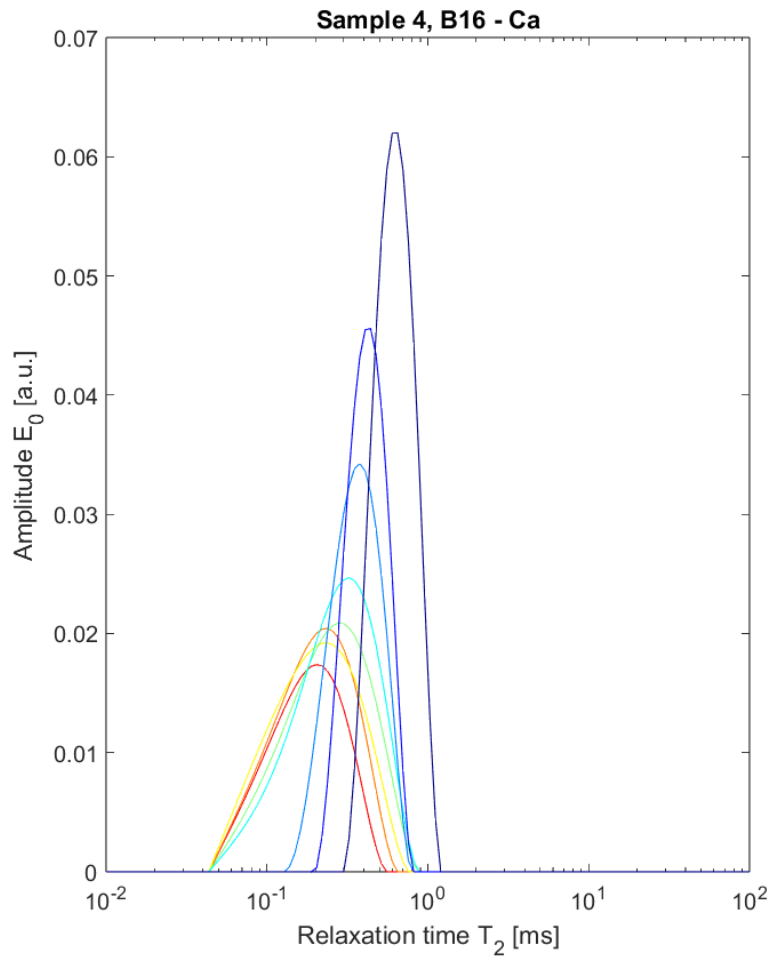
- Further laboratory experiments to help interpreting relaxation times of clay rocks:
- Different well-defined clay mineral powders + grinded OPALINUS clay samples
- Drying and wetting circles in a climate chamber => NMR at each moisture step
- **Most important question: Do we see the interlayer water in the Opalinus clay?**
- **...and can we quantify the content using NMR relaxometry?**

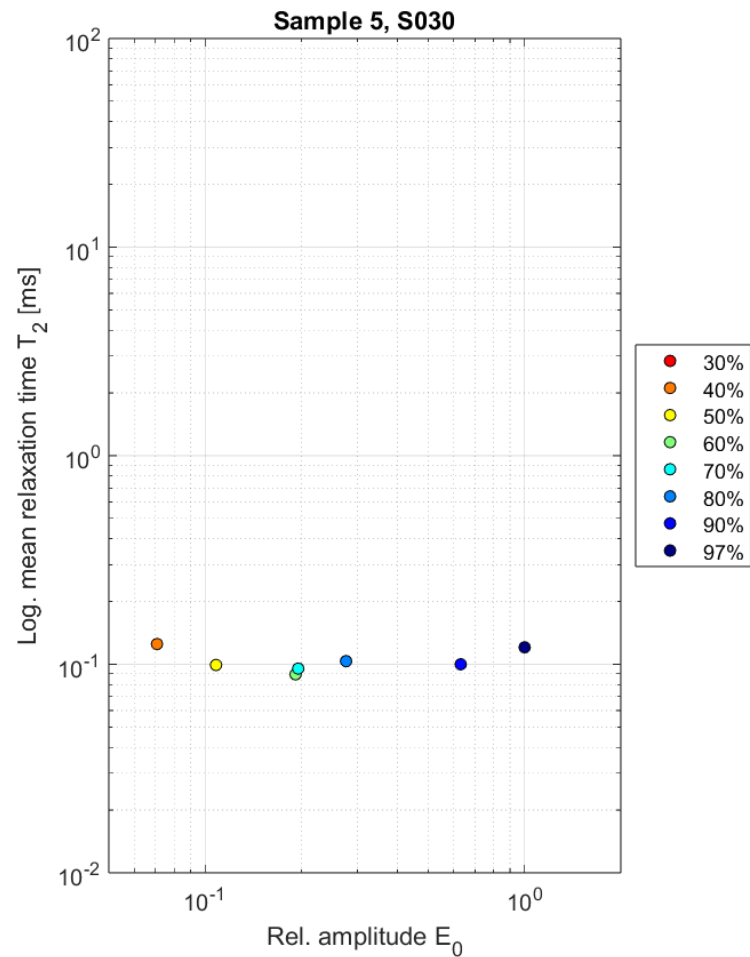
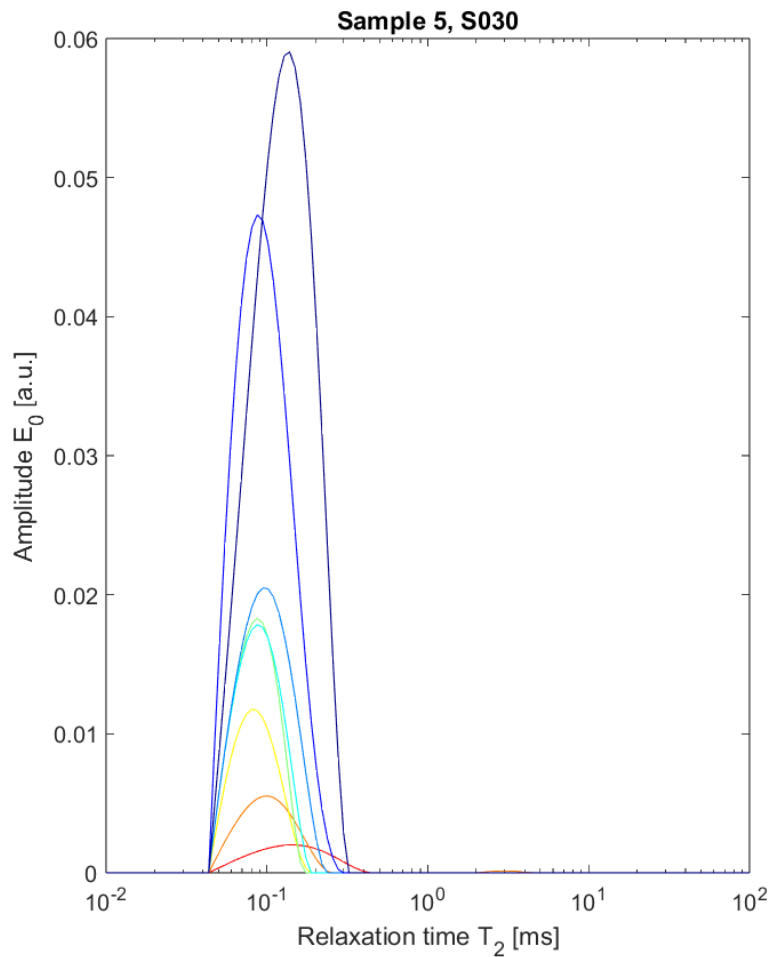


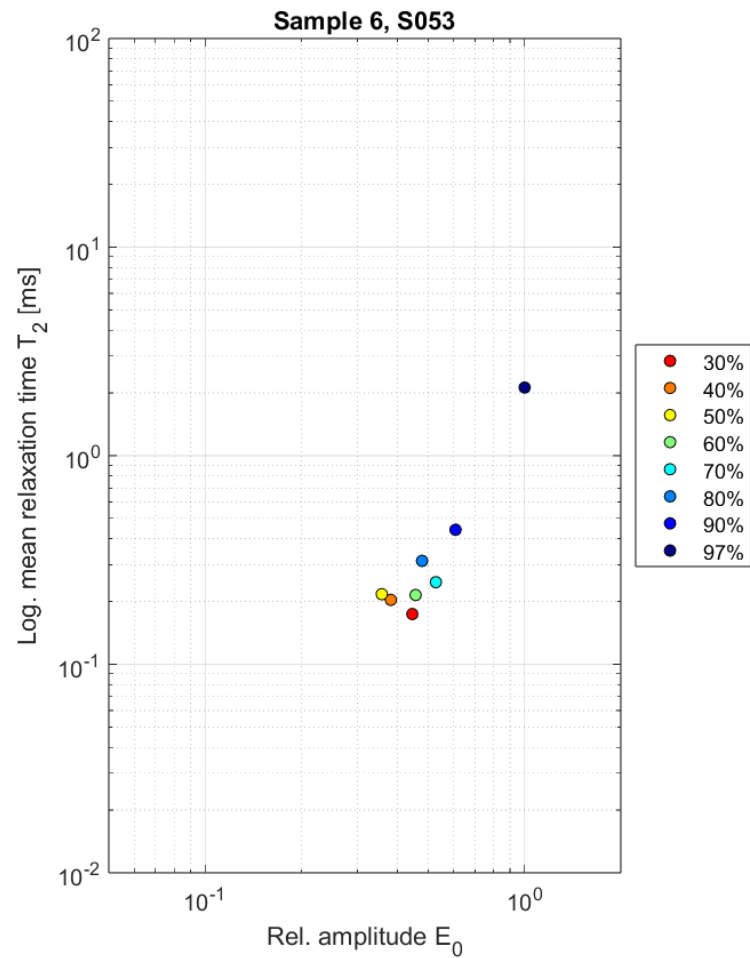
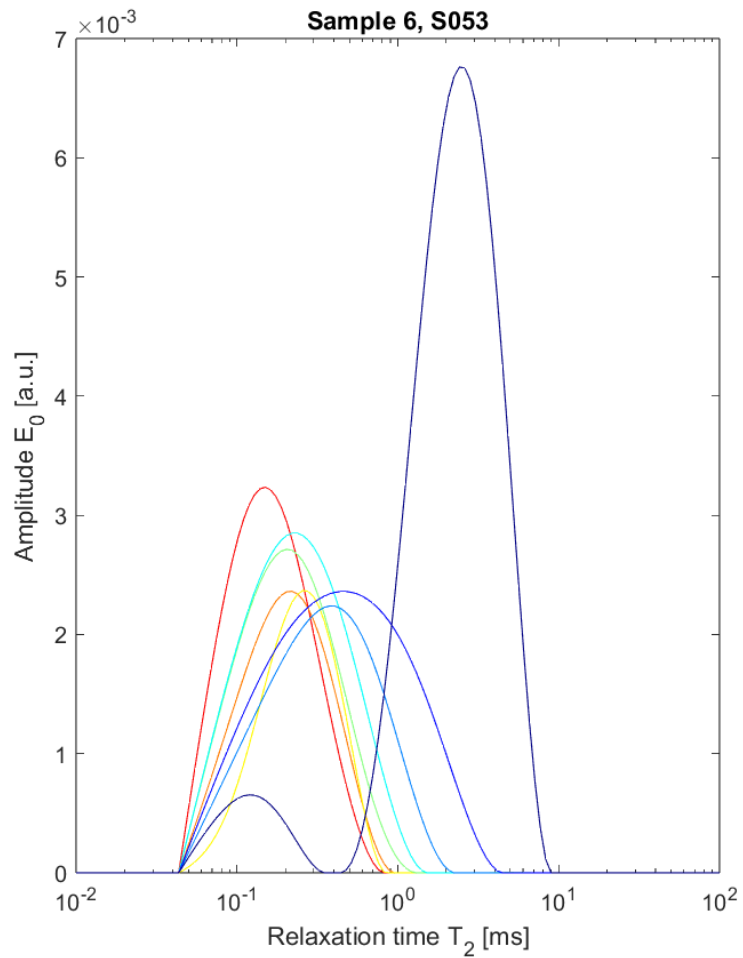


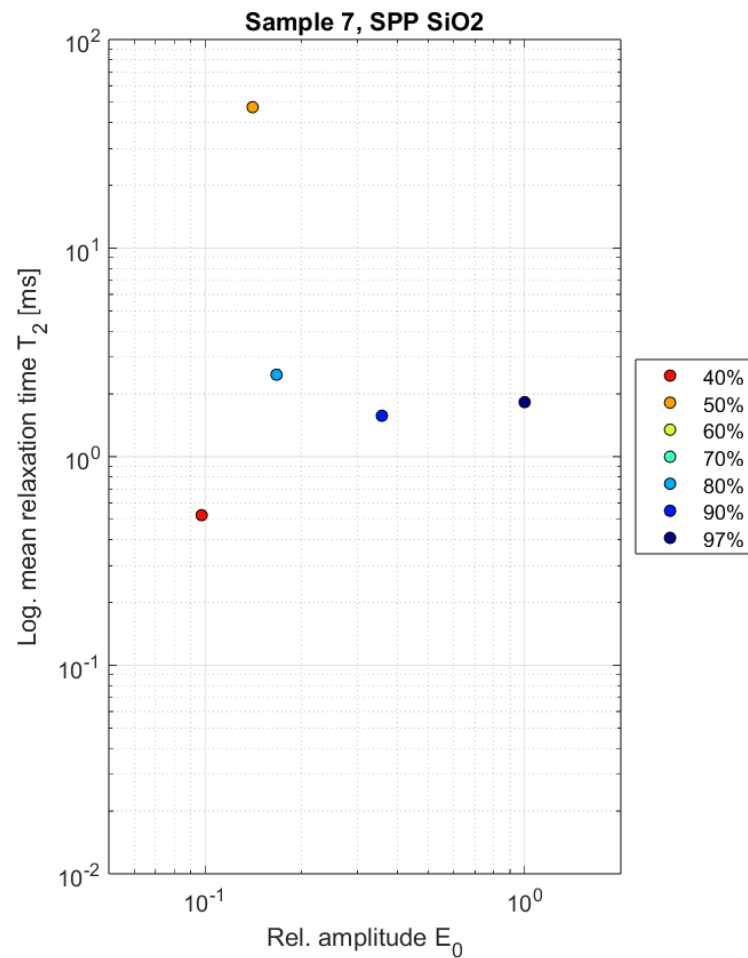
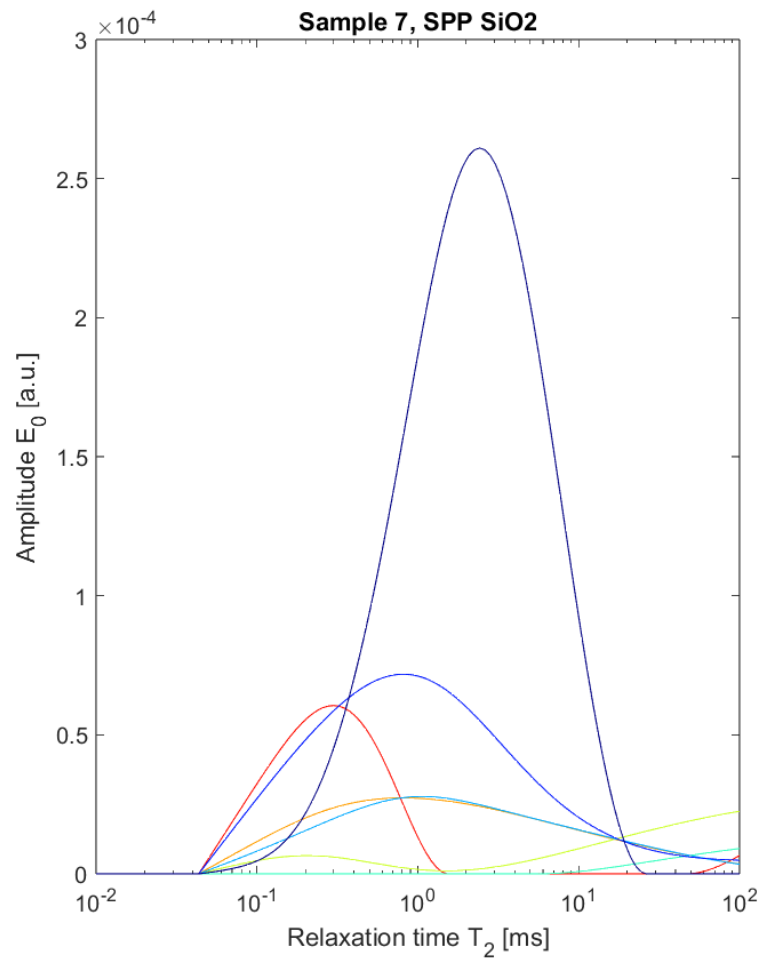


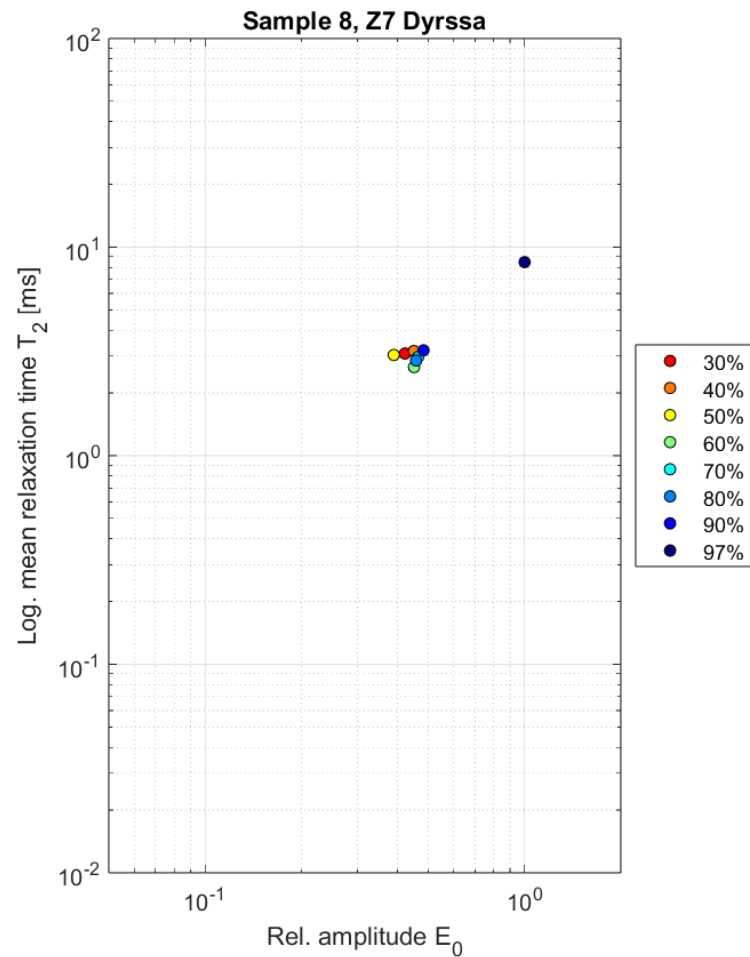
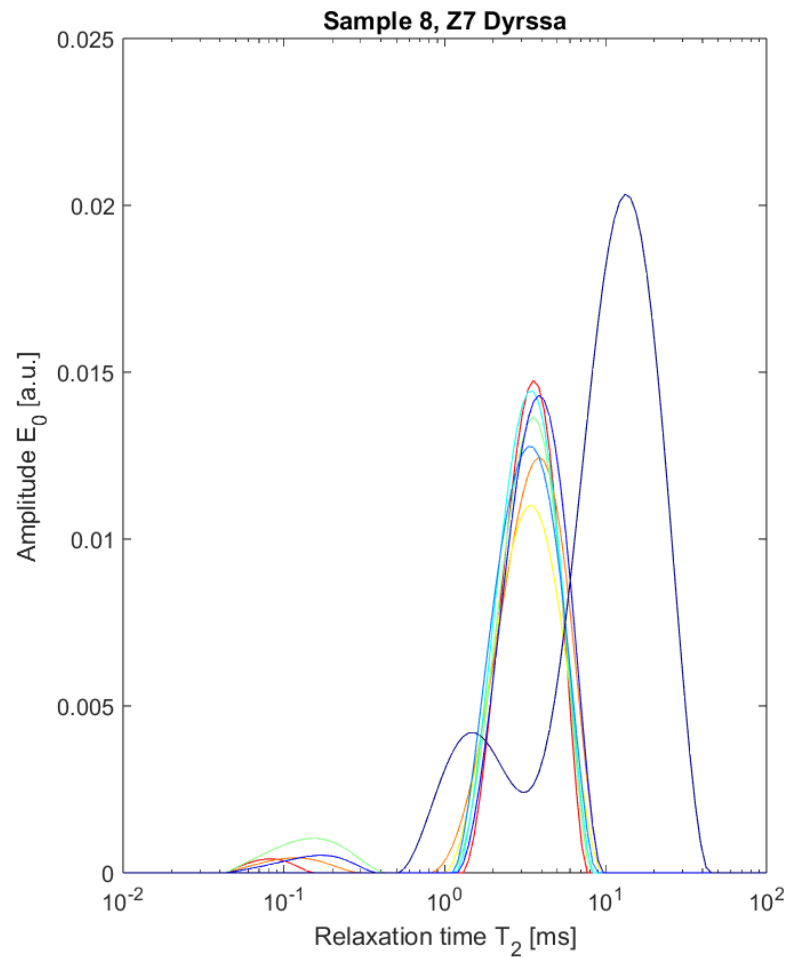


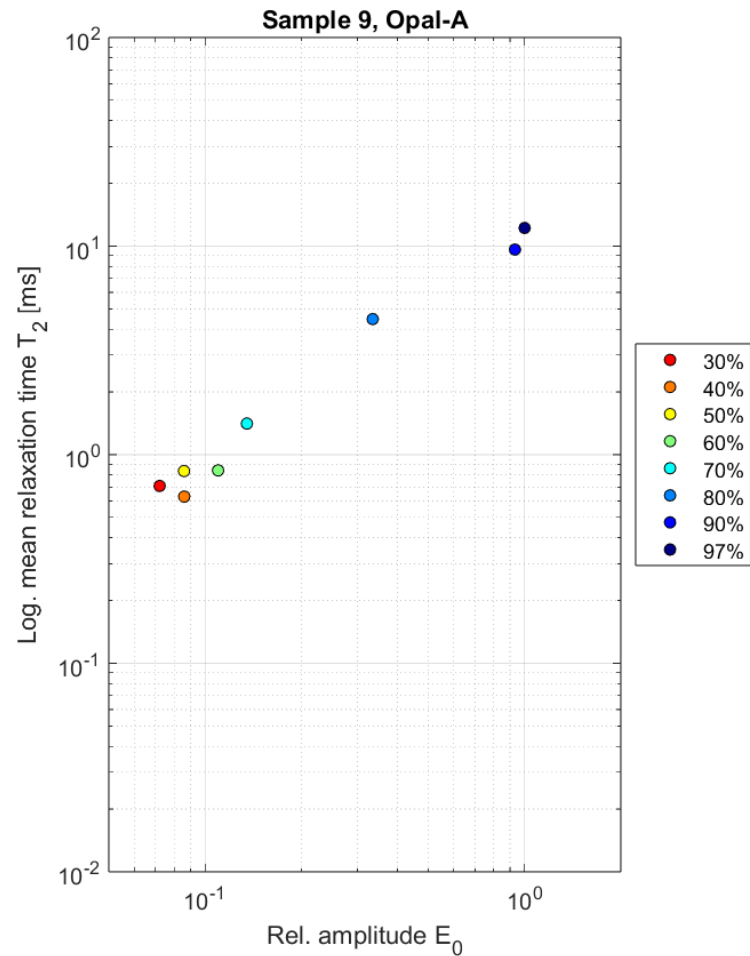
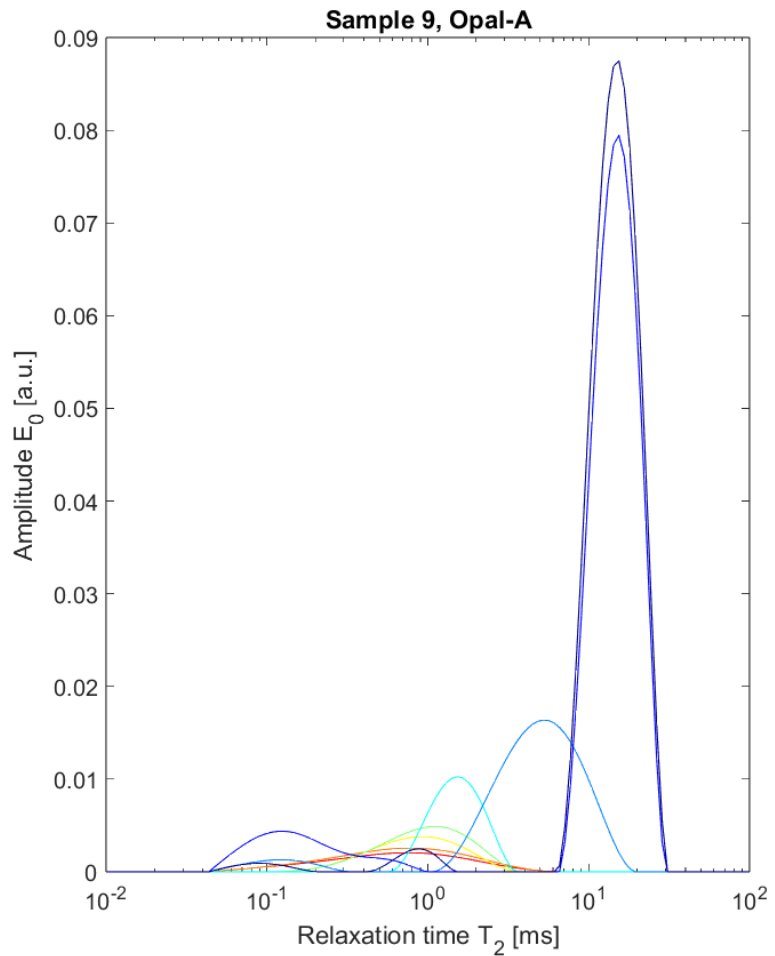




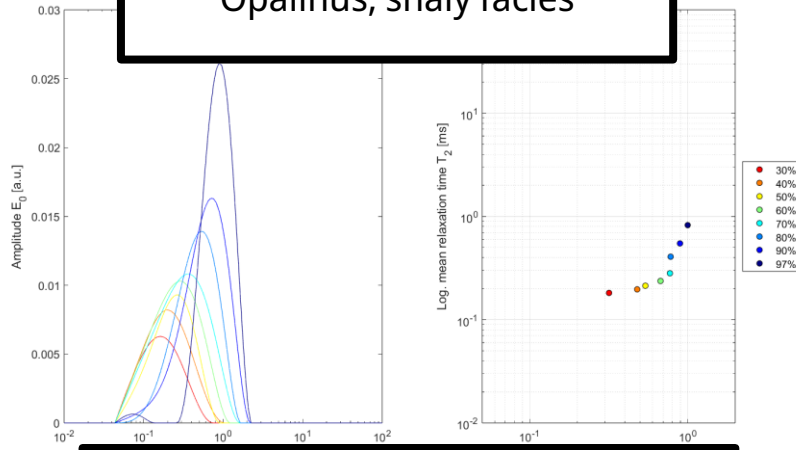




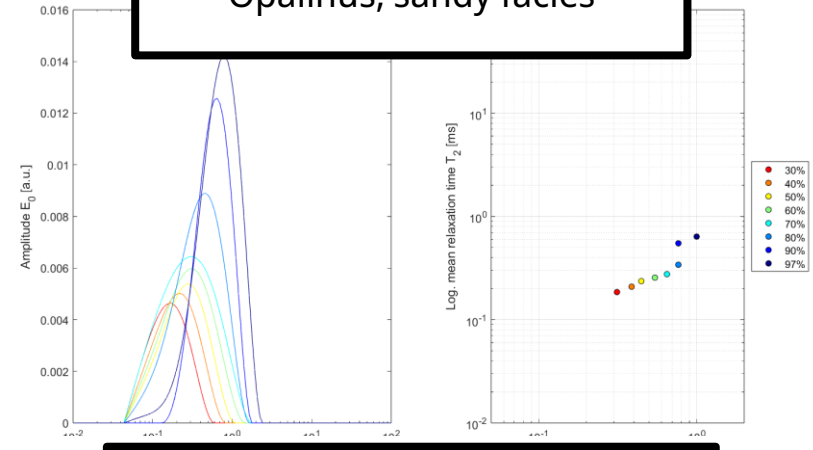




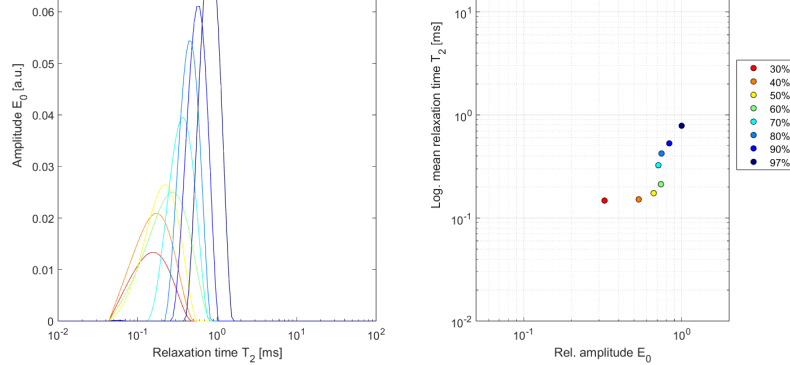
Opalinus, shaly facies



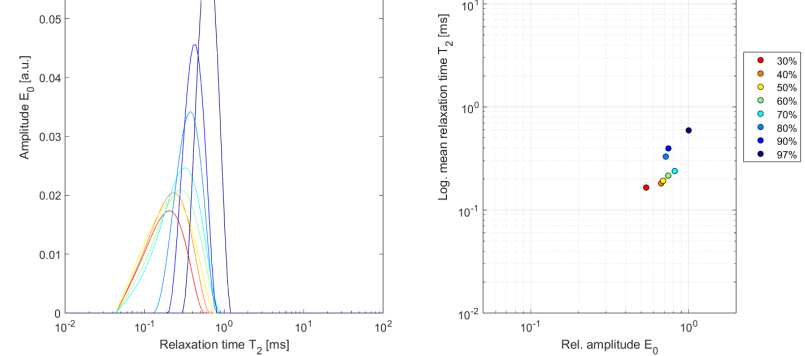
Opalinus, sandy facies



Na-Bentonite (++ interlayer water)

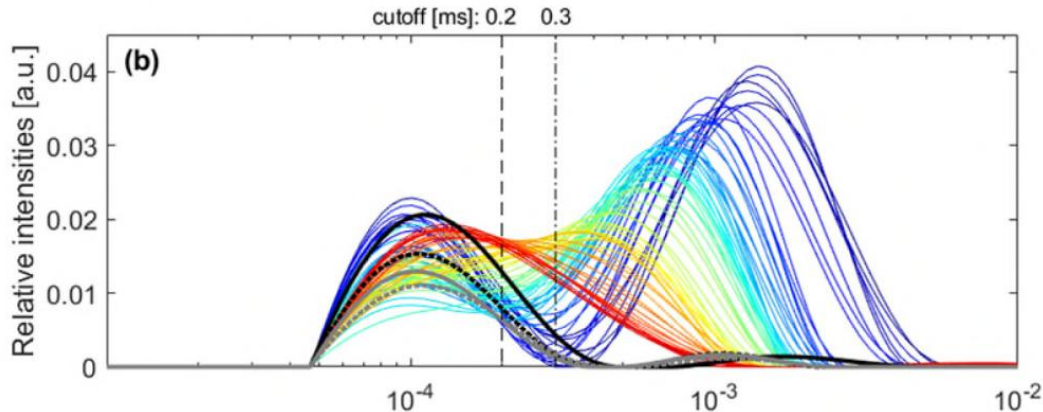


Ca-Bentonite (- interlayer water)



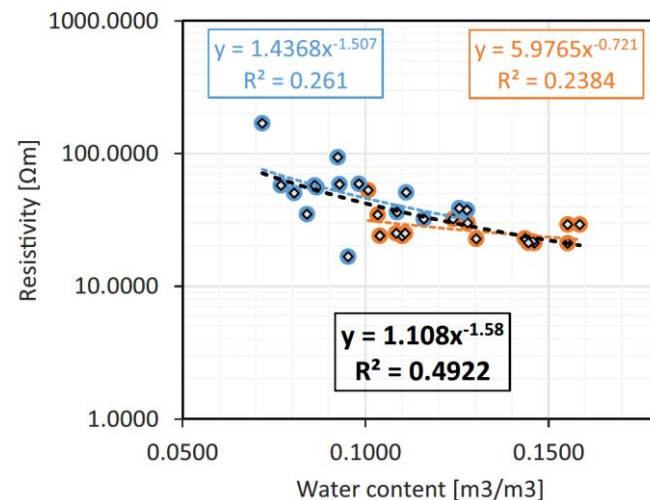
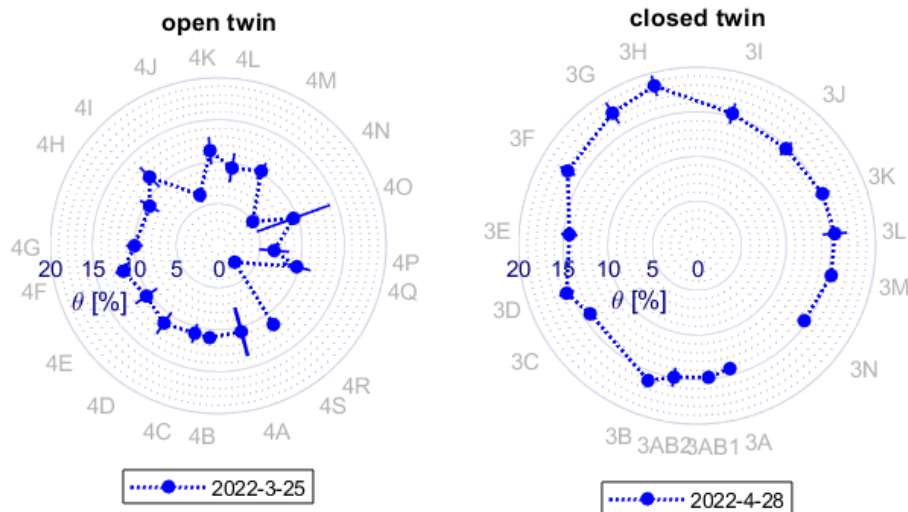
Preliminary results

- NMR response of Opalinus powders very similar to Bentonite, esp. Na-Bentonite powder
- Cut-off values for interlayer water found to be at ~0.5 ms to 0.6 ms
- Opa in natural solid formation: bimodal NMR response indicates the two types of water binding
- Separating them using cut-off values seem possible, however, diffusion and data noise smears the water signal in the range of the cut-off => **quantification still challenging!**

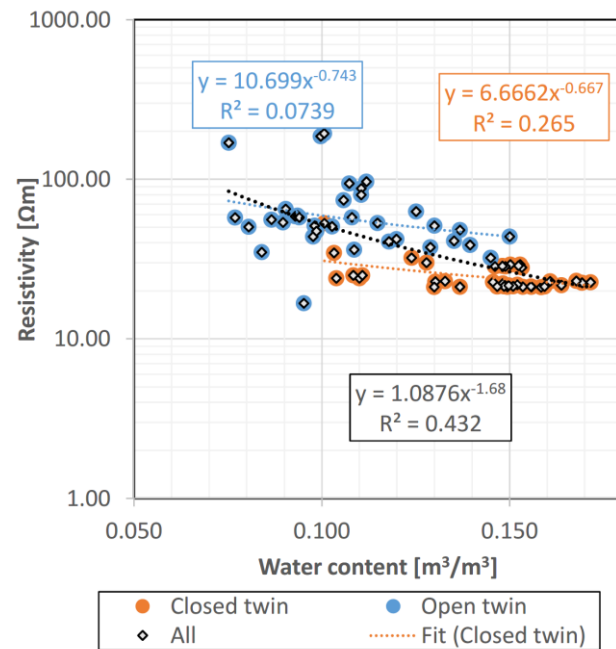


**Thank you
for your attention!**

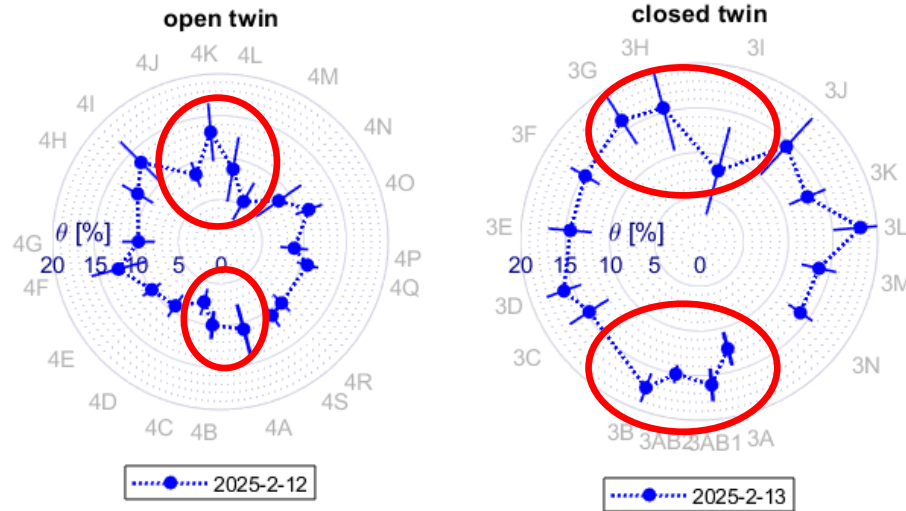
Seasonal measurements in both niches - data until Feb 2025



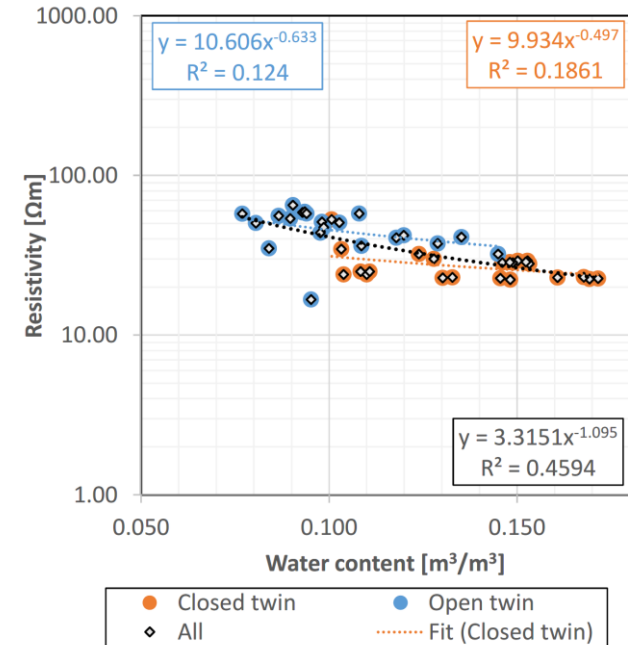
- Closed twin
- Open twin
- ◇ All
- Fit (Closed niche)
- Fit (Open niche)
- Fit (All)



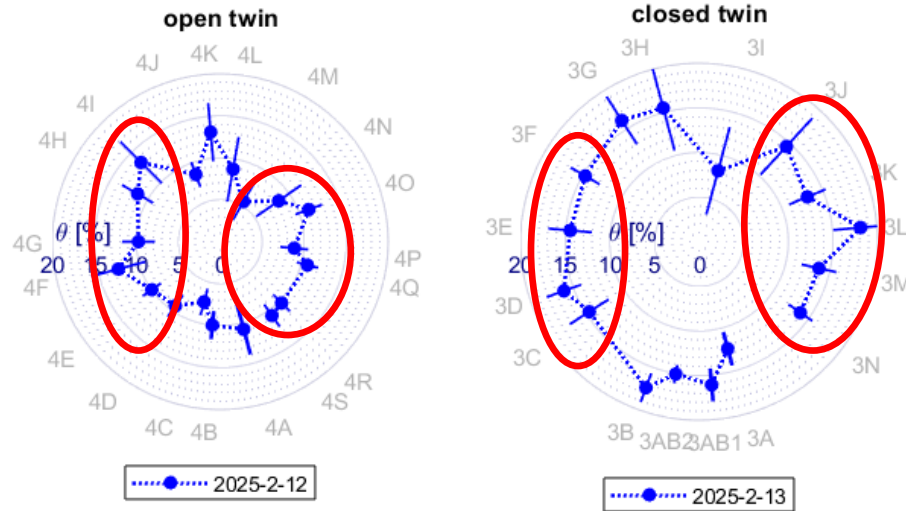
Seasonal measurements in both niches - correlation with ERT



Only top and bottom



Seasonal measurements in both niches - correlation with ERT



Only sidewalls

